

Rosen — §2.4: Sequences and Summations

Selected Exercises

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Exercise 11

Let $a_n = 2^n + 5 \cdot 3^n$ for $n = 0, 1, 2, \dots$.

- (a) Find a_0, a_1, a_2, a_3 , and a_4 .
- (b) Show that $a_2 = 5a_1 - 6a_0$, $a_3 = 5a_2 - 6a_1$, and $a_4 = 5a_3 - 6a_2$.
- (c) Show that $a_n = 5a_{n-1} - 6a_{n-2}$ for all integers n with $n \geq 2$.

Exercise 27

Show that if a_n denotes the n th positive integer that is not a perfect square, then $a_n = n + \{\sqrt{n}\}$, where $\{x\}$ denotes the integer closest to the real number x .

Exercise 35

Show that

$$\sum_{j=1}^n (a_j - a_{j-1}) = a_n - a_0,$$

where $a_0, a_1, \dots, a_n$ is a sequence of real numbers. This type of sum is called *telescoping*.

Exercise 37

Sum both sides of the identity $k^2 - (k-1)^2 = 2k - 1$ from $k = 1$ to $k = n$ and use Exercise 35 to find

- (a) a formula for $\sum_{k=1}^n (2k - 1)$ (the sum of the first n odd natural numbers).
- (b) a formula for $\sum_{k=1}^n k$.

Exercise 38

Use the technique given in Exercise 35, together with the result of Exercise 37(b), to derive the formula for $\sum_{k=1}^n k^2$ given in Table 2. [Hint: Take $a_k = k^3$ in the telescoping sum in Exercise 35.]

Exercise 41

Find a formula for $\sum_{k=0}^m \lfloor \sqrt{k} \rfloor$, when m is a positive integer.

Exercise 42

Find a formula for $\sum_{k=0}^m \lfloor \sqrt[m]{k} \rfloor$, when m is a positive integer.